

CSS

Cascading Style Sheet

*It is a language that is used to describe the **style** of a document.*

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Basic Syntax

Selector

```
h1 {  
  color: red;  
}
```

Property

Value

The diagram illustrates the basic syntax of a CSS rule. It shows a selector 'h1' followed by a block of properties and values enclosed in curly braces. An arrow points from the label 'Selector' to the 'h1' text. Another arrow points from the label 'Property' to the 'color:' text. A third arrow points from the label 'Value' to the 'red' text. The entire code snippet is: `h1 {
 color: red;
}`

Including Style

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- Inline

```
<h1 style="color: red"> Apna College </h1>
```

- <style> tag

```
<style>  
  h1 {  
    color : red;  
  }  
</style>
```

Including Style

- External Stylesheet

Writing CSS in a separate document & linking it with HTML file

Color Property

Used to set the color of foreground

color: red;

color: pink;

color: blue;

color: green;

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Background Color Property

Used to set the color of background

background-color: red;

background-color: pink;

background-color: blue;

background-color: green;

Including Style

- External Stylesheet

Writing CSS in a separate document & linking it with HTML file

Color Property

Used to set the color of *foreground*

```
color: red;
```

```
color: pink;
```

```
color: blue;
```

```
color: green;
```

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Background Color Property

Used to set the color of background

background-color: red;

background-color: pink;

background-color: blue;

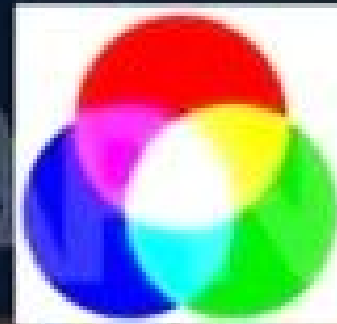
background-color: green;

Color Systems

- RGB

color: `rgb(255, 0, 0);`

color: `rgb(0, 255, 0);`



Color Systems

- Hex (Hexadecimal)

color: #ff0000;

color: #00ff00;

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Selectors

- Universal Selector

```
* { }
```

- Element Selector

```
h1 { }
```

- Id Selector

```
#myId { }
```

- Class Selector

```
.myClass { }
```

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Practice Set 1

Q1: Create a simple div with an id "box".

Add some text content inside the div.

Set its background color to blue.

Q2: Create 3 headings with h1, h2 & h3.

Give them all a class "heading" & set color of "heading" to red.

Q3: Create a button & set its background color to :

- green using css stylesheet
- blue using <style> tag
- pink using inline style

Text Properties

text-align

text-align : left / right / center

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Text Properties

text-decoration

text-decoration : underline / overline / line-through

Practice Set 1



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Text Properties

text-decoration

text-decoration : underline / overline / line-through

Text Properties

font-weight

font-weight : normal / bold / bolder / lighter

font-weight : 100-900

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Text Properties

font-family

font-family : arial

font-family : arial, roboto

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- **Generic font families:** Generic font families are determined by font family properties such as serifs – which are decorative strokes on the ends of letters – or cursive strokes. The generic font family name will specify the attribute that all fonts within that family share, like serif, sans-serif, or monospace.
- **Specific font families:** Specific font families are specific fonts with different styles within the one font family name, such as Arial, Times New Roman, and Tahoma.

5 Generic Font Families

Here is an overview of the generic font families found in many word processing programs:

1. **Serif:** Serif fonts are traditional typefaces using characters that have serifs which are small winged or flared tips extending off the top of a letter. Serif fonts are typically used in printed books, newspapers, and magazines. Some popular serif fonts include Times New Roman, Garamond, Palatino, and Georgia.
2. **Sans-Serif:** Sans-serif fonts use characters without serifs and are more commonly seen in digital formats. A sans-serif font will typically be the default font in digital word processing programs. Sans-serif fonts include Arial, Helvetica, Verdana, Trebuchet MS, and Gill Sans.
3. **Cursive:** Cursive fonts use characters that have connective strokes which give the font a handwritten appearance. Cursive fonts include Comic Sans MS, Adobe Postica, Scripto, and Zapf-Chancery.
4. **Fantasy:** Fantasy fonts are stylized fonts that still maintain the characteristics of non-cursive, traditional alphabet glyphs. Examples include Cottonwood, Gribble, and Alpha Geometrique.
5. **Monospace:** Fonts in the monospace font family have characters that are all the same width, giving text the appearance of a manual monospaced typewriter. Examples of monospaced fonts include Courier New, Monaco, Lucida Console, Consolas, and Everson Mono.

Units in CSS

Absolute

pixels (px)

96px = 1 inch

font-size: 2px;

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Text Properties

line-height

line-height : 2px

line-height : 3

line-height : normal

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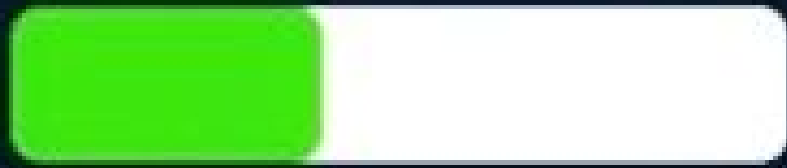
Text Properties

text-transform

text-transform : uppercase / lowercase / capitalize / none

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Level 2

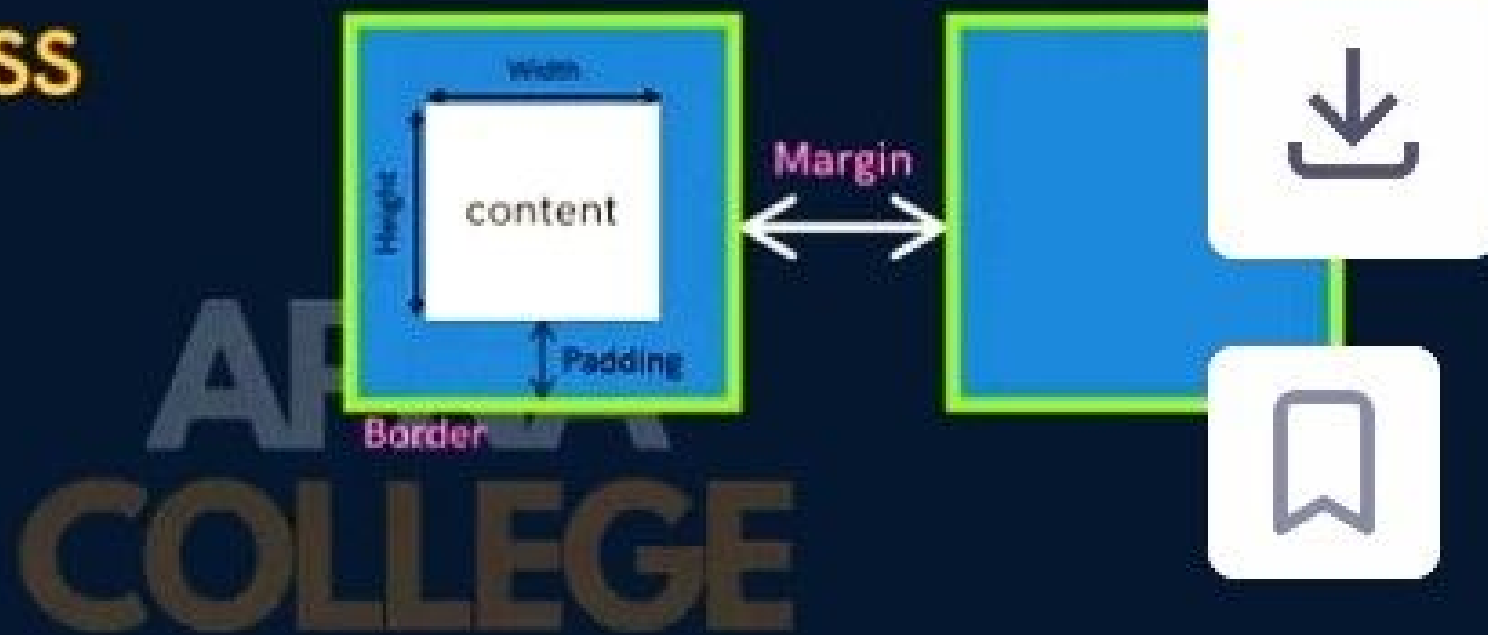


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Box Model in CSS

- Height
- Width
- Border
- Padding
- Margin



Practice Set 2



Q1: Create a heading centred on the page with all of its text capitalized by default.

Q2: Set the font family of all the content in the document to "Times New Roman".



Q3: Create one div inside another div.

Set id & text "outer" for the first one & "inner" for the second one.

Set the outer div text size to 25px & inner div text size to 10px.

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Width

By default, it sets the content area **width** of the element



```
div {  
  width: 50px;  
}
```

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Border

Used to set an element's **border**

border-width : 2px;
border-style : solid / dotted / dashed
border-color : black;



Border

Shorthand

border : 2px solid black;

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Border

Used to **round the corners** of an element's outer border edge

border-radius : 10px;

border-radius : 50%;

Padding

- padding-left
- padding-right
- padding-top
- padding-bottom



Padding

Shorthand

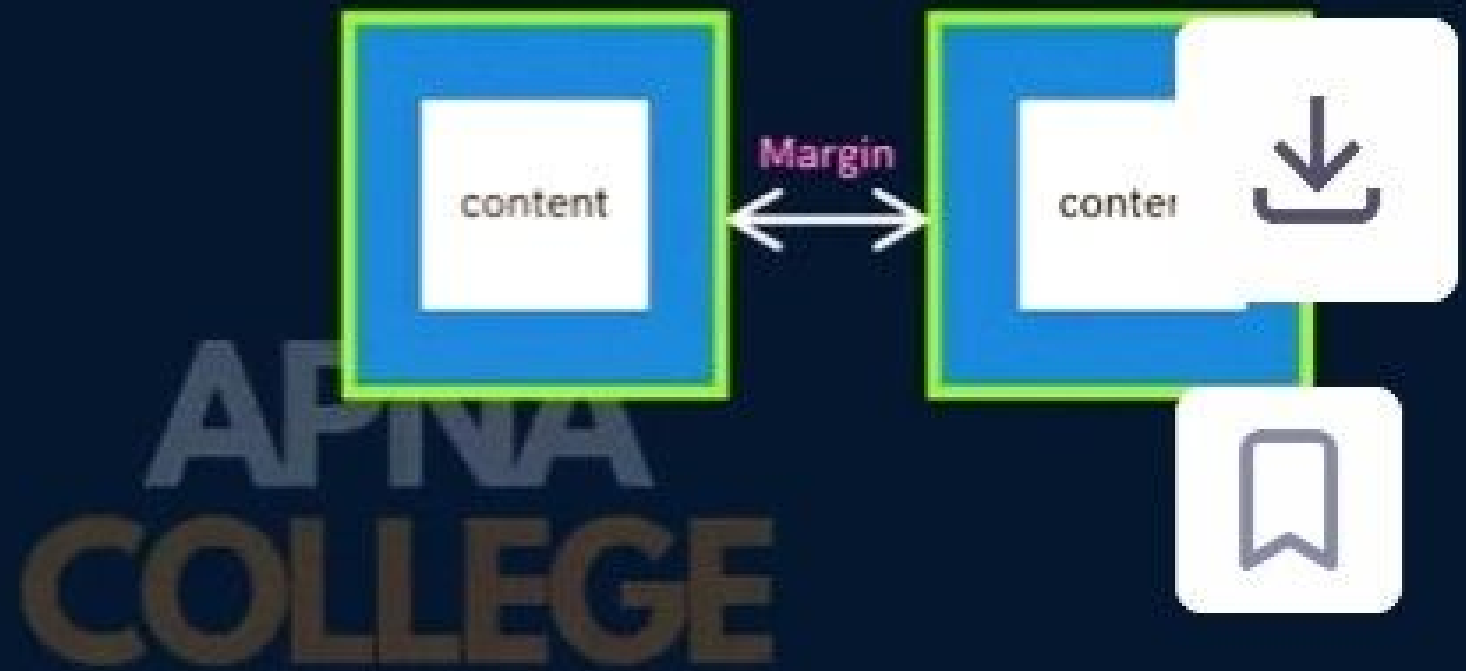
padding: 50px;

padding: 1px 2px 3px 4px;

top | right | bottom | left -> clockwise

Margin

- margin-right
- margin-left
- margin-top
- margin-bottom



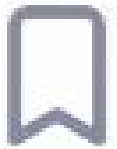
Margin

Shorthand

margin: 50px;

margin: 1px 2px 3px 4px;

top | right | bottom | left -> clockwise



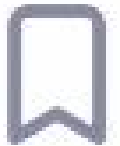
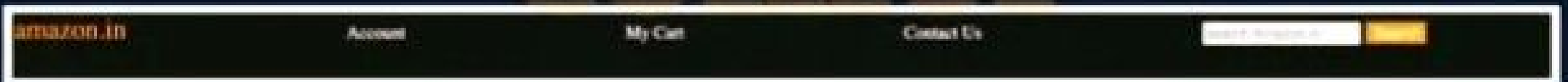
Practice Set 3



Q1: Create a div with height & width of 100px.

Set its background color to green & the border radius to 50%.

Q2: Create the following navbar.



Display Property

display: inline / block / inline-block / none

- **inline** - Takes only the space required by the element. (no margin, padding)
- **block** - Takes full space available in width.
- **inline-block** - Similar to inline but we can set margin & padding.
- **none** - To remove element from document flow.

Visibility

visibility: hidden;

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Note : When visibility is set to none, space for the element is reserved.

But for display set to none, no space is reserved or blocked for the element.

Alpha Channel

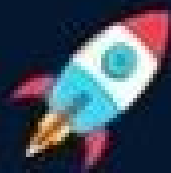
opacity (0 to 1)

- RGBA

color: `rgba(255, 0, 0, 0.5);`

color: `rgba(255, 0, 0, 1);`

Practice Set 4



- Q1: Create a webpage layout with a header, a footer & a content area containing
Set the height & width of divs to 100px.
(add the previous navbar in the header)
- Q2: Add borders to all the divs.
- Q3: Add a different background color to each div with an opacity of 0.5
- Q4: Give the content area an appropriate height.

Level 3



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Units in CSS

Relative

%

em

rem

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Percentage (%)

It is often used to define a size as relative to an element's parent object.

width : 33% ;

margin-left : 50% ;

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Em

Unit	Relative to
em	Font size of the parent, in the case of typographical properties like <code>font-size</code> , and font size of the element itself, in the case of other properties like <code>width</code> .

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Rem (Root Em)

Unit	Relative to
rem	Font size of the root element.

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Others

vh: relative to 1% viewport height

vw : relative to 1% viewport width

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Position

The position CSS property sets **how an element is positioned** in a document.

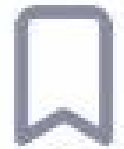
position : static / relative / absolute / fixed

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Position



- **static** - default position (The top, right, bottom, left, and z-index properties have no effect)
- **relative** - element is relative to itself. (The top, right, bottom, left, and z-index properties have an effect)
- **absolute** - positioned relative to its closest positioned ancestor. (removed from the flow)
- **fixed** - positioned relative to browser. (removed from flow)
- **sticky** - positioned based on user's scroll position



z-index

It decides the **stack level** of elements

Overlapping elements with a larger z-index cover those with a smaller one.

z-index : auto (0)

z-index : 1 / 2 / ...

z-index : -1 / -2 / ...

Background Image

Used to set an image as background

```
background-image : url("image.jpeg");
```

Background Size

background-size : cover / contain / auto

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Practice Set 5



Qs: Create the following layout using the given html.

- Give the div a height, width & some background image.
- Use the appropriate position property for the div element to place it at the right end of the page. (The div should not move even on scroll)
- Use z-index to place the div on top of page.

```
<p> lorem*5 </p>
```

```
<div> Love Nature </div>
```

```
<p> lorem*5 </p>
```

Level 4



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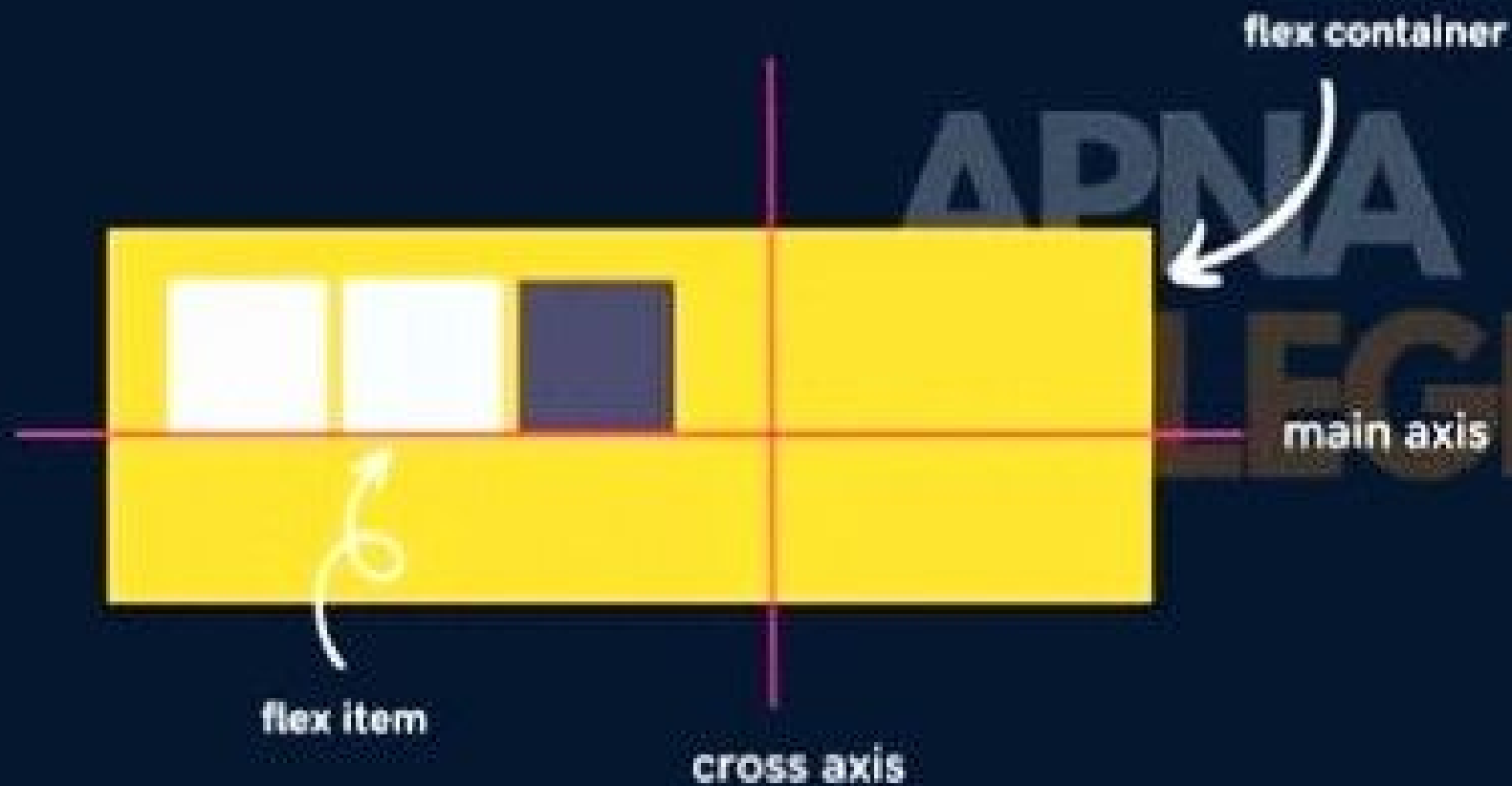
Flexbox

Flexible Box Layout

It is a one-dimensional layout method for arranging items in rows or columns.

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The Flex Model



Flexbox Direction

It sets how flex items are placed in the flex container, along which axis and direction

- flex-direction : row; (default)
- flex-direction : row-reverse;
- flex-direction : column;
- flex-direction : column-reverse;

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Flex Properties

for Flex Container

- `justify-content` : alignment along the main axis.
`flex-start / flex-end / centre / space-evenly /`
- `flex-wrap` : `nowrap / wrap / wrap-reverse`
- `align-items` : alignment along the cross axis.
- `align-content` : alignment of space between & around the content along cross-axis

Flex Properties

for Flex Item

- **align-self** : alignment of individual along the cross axis.
- **flex-grow** : how much a flex item will grow relative to the rest of the flex items space is available
- **flex-shrink** : how much a flex item will shrink relative to the rest of the flex items if space is available

Practice Set 6



Qs: Create a navbar with 4 options in the form of anchor tags inside list items.
Now, use flexbox to place them all spaced equally in a single line.

Qs: Use flexbox to center one **div** inside another **div**.

Qs: Which has higher priority - align-items or align-self?

Media Queries

Help create a responsive website

```
@media (width : 600px) {  
  div {  
    background-color : red;  
  }  
}
```

```
@media (min-width : 600px) {  
  div {  
    background-color : red;  
  }  
}
```

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Media Queries

```
@media (min-width : 200px) and (min-width : 300px) {  
  div {  
    background-color : red;  
  }  
}
```

Practice Set 7



Qs: Add a media query to implement the following:

- the color of a div changes to green for viewport width less than 300px
- the color of a div changes to pink for width between 300px & 400px
- the color of a div changes to red for width between 400px & 600px
- the color of a div changes to blue for width above 600px

Level 5



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Transitions

Transitions enable you to define the transition between two states of an element

- `transition-property` : property you want to transition (font-size, width etc.)
- `transition-duration` : 2s / 4ms ..
- `transition-timing-function` : ease-in / ease-out / linear / steps ..
- `transition-delay` : 2s / 4ms ..

Transition Shorthand

property name | duration | timing-function | delay

transition: font-size 2s ease-in-out 0.2s;

CSS Transform

Used to apply 2D & 3D transformations to an element

- rotate

transform: rotate(45deg);

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CSS Transform

- scale

transform: scale(2);

transform: scale(0.5);

transform: scale(1, 2);

transform: scaleX(0.5);

transform: scaleY(0.5);

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CSS Transform

- translate

transform: translate(20px);

transform: translate(20px, 50px);

transform: translateX(20px);

transform: translateY(20px);

CSS Transform

- skew

transform: skew (30deg);

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Animation

To animate CSS elements

```
@keyframe myName {  
  from { font-size : 20px; }  
  to { font-size : 40px; }  
}
```

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Animation Properties

- animation-name
- animation-duration
- animation-timing-function
- animation-delay
- animation-iteration-count
- animation-direction

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Animation Shorthand

animation : myName 2s linear 3s infinite normal

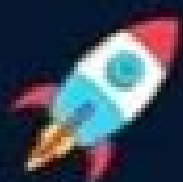
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% in Animation

```
@keyframe myName {  
  0% { font-size : 20px; }  
  50% { font-size : 30px; }  
  100% { font-size : 40px; }  
}
```

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Practice Set 8



Qs: Create a simple loader using CSS

Step1 : create a div with circular shape & a thick border from one end (top/bottom/left/right)

Step2 : To make it spin create an animation which transforms it from 0deg to

Step3 : Add the animation property to the loader with infinite duration